Day 3 – Agile and Scrum Foundations

Topics covered: Agile Overview · Scrum Roles, Events and Artifacts · User Stories, Backlog and Estimation

Set 2 – Insurance Claims Portal

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Business Scenario	An insurer is building a self-service portal where customers lodge claims, upload documents and track settlement status.
Assumptions	The product is called **ClaimEase**. The name is chosen for this exercise. No team details were supplied, so roles are described by title (Product Owner, Scrum Master, Developers) and not by person. The users are **policyholders** (customers) and **claims officers** (insurer staff). Policyholders already have a portal login. Story points use the Planning Poker values 1, 2, 3, 5, 8 and 13, agreed by the Developers. Story IDs (CLM-101 and so on) are the team's own IDs. Jira numbers issues automatically, so the team ID is written at the start of each story title.

Task 1 – Product vision and Waterfall versus Agile

Product vision statement

> **For** policyholders **who** need to make an insurance claim after a loss and are tired of paperwork, phone calls and not knowing what is happening, **ClaimEase** is a self-service claims portal **that** lets them lodge a claim, upload documents and follow the settlement online.
>
> **Unlike** the current paper and call-centre process, **our product** shows a clear status at every step and tells the customer as soon as the claim is settled.

Waterfall versus Agile

Aspect	Waterfall	Agile (Scrum)
Requirements	Fixed in full before building starts.	Start with a backlog and refine it every sprint.
Delivery	One release at the end of the project.	A working increment every 2 weeks.
Customer feedback	Late, mostly at final testing.	At every Sprint Review.
Change	Slow, through formal change control.	Welcome. The Product Owner reorders the backlog.
Risk	Problems found late, when they cost most.	Found early, in small steps.

Why Agile suits this programme:

- Claim rules, required documents and regulations change, so the requirements cannot be fixed on day one.
- Claims officers and policyholders can see working screens early and correct them, instead of finding gaps at the end.
- Value comes early: claim registration can go live first, and tracking and notifications follow in later sprints.

Task 2 – Four epics and 13 user stories

Epic	Name	Goal
E1	Claim Registration	A policyholder can lodge a valid claim and get a reference number.

Epic	Name	Goal
E2	Document Upload	Supporting documents can be uploaded, checked and verified.
E3	Claim Tracking	Customers and officers can see and update where a claim stands.
E4	Settlement Notifications	Customers are told the decision and the settlement details.

ID	Epic	User story (first draft)
CLM-101	E1	As a policyholder, I want to select the policy I am claiming against, so that my claim is linked to the correct policy.
CLM-102	E1	As a policyholder, I want to enter the incident details (date, place, claim type and description), so that the insurer knows what happened.
CLM-103	E1	As a policyholder, I want the claim form to be easy to use and to validate my entries, so that I make no mistakes.
CLM-104	E1	As a policyholder, I want to submit my claim and receive a claim reference number, so that I can refer to it later.
CLM-201	E2	As a policyholder, I want to upload supporting documents (photos, PDFs and videos) with virus scanning, preview and secure storage, so that the insurer can verify my loss.
CLM-202	E2	As a policyholder, I want to see which documents my claim type needs and which are still missing, so that I know what to provide.
CLM-203	E2	As a claims officer, I want to view each uploaded document and mark it accepted or rejected with a reason, so that the claim can move forward.
CLM-301	E3	As a policyholder, I want to see a list of all my claims with their current status, so that I know where each claim stands.
CLM-302	E3	As a policyholder, I want to see the dated status history of one claim, so that I understand what has happened so far.
CLM-303	E3	As a claims officer, I want to update the status of a claim, so that the policyholder sees the latest progress.
CLM-401	E4	As a policyholder, I want to receive an SMS and an e-mail when my claim is approved or rejected, so that I do not have to keep checking the portal.
CLM-402	E4	As a policyholder, I want to see the settlement amount and the expected payment date on my claim, so that I can plan my finances.
CLM-403	E4	As a claims officer, I want to calculate the settlement, approve it, pay it into the customer's bank account and notify the customer, so that the claim is closed.

Task 3 – INVEST review and vertical splitting

ID	I	N	V	E	S	T	Result
CLM-101	✔	✔	✔	✔	✔	✔	Passes all six criteria.
CLM-102	✔	✔	✔	✔	✔	✔	Passes all six criteria.
CLM-103	✔	✔	✔	✔	✔	✘	**Fails T.** "Easy to use" cannot be tested.
CLM-104	✔	✔	✔	✔	✔	✔	Passes all six criteria.
CLM-201	✔	✔	✔	✔	✘	✔	**Fails S.** Three file types, virus scanning, preview and storage are too much for one story.

ID	I	N	V	E	S	T	Result
CLM-202	✔	✔	✔	✔	✔	✔	Passes all six criteria.
CLM-203	✔	✔	✔	✔	✔	✔	Passes all six criteria.
CLM-301	✔	✔	✔	✔	✔	✔	Passes all six criteria.
CLM-302	✔	✔	✔	✔	✔	✔	Passes all six criteria.
CLM-303	✔	✔	✔	✔	✔	✔	Passes all six criteria.
CLM-401	✔	✔	✔	✔	✔	✔	Passes all six criteria.
CLM-402	✔	✔	✔	✔	✔	✔	Passes all six criteria.
CLM-403	✔	✔	✔	✘	✘	✔	**Fails E and S.** Four jobs in one story, and the bank payment link is not yet known.

Stories too large: **CLM-201** and **CLM-403**. CLM-103 is also rewritten because it cannot be tested.

Why the split is vertical and not by technical layer

A **horizontal** split cuts the story by technical layer (database, then API, then screen). None of the pieces gives the user anything, and nothing can be demonstrated until all layers are done. A **vertical** split cuts a thin slice through every layer (screen, logic and data), so each slice can be built, tested and shown to the user on its own.

Story	Wrong: split by layer	Right: vertical slices
CLM-201	Create upload database tables Build upload API Build upload screen	**CLM-201** Upload a photo (JPG or PNG, up to 5 MB) and see it in the claim's document list. **CLM-204** Upload a PDF (up to 10 MB) in the same way. **CLM-205** An infected file is blocked and the customer sees a message. Videos and preview are parked for a later release.
CLM-403	Create settlement database tables Build payment API Build settlement screen	**CLM-403** Approve or reject a claim and record the settlement amount, so the decision shows on the claim. **CLM-404** Pay the approved amount to the customer's registered bank account and mark the claim Settled.

CLM-103 rewritten: As a policyholder, I want the claim form to show a clear error next to any missing or invalid field (for example, an incident date in the future), so that I can correct it before submitting.

The backlog now has **16 stories**: the 13 drafts, with CLM-201 and CLM-403 rewritten, plus the three new slices CLM-204, CLM-205 and CLM-404.

Task 4 – Acceptance criteria for the Claim Registration stories

Scenarios marked **(Negative)** cover failures. The validation failure scenarios are CLM-102 scenario 2 and CLM-103 scenarios 1 and 2.

CLM-101 – Select the policy for the claim

Scenario	Given	When	Then
1. Policy selected	A logged-in policyholder has two active policies	They open Lodge a claim	Both policies are listed with number and type, and the customer can select one to continue.

Scenario	Given	When	Then
2. No active policy (Negative)	A logged-in policyholder has no active policy	They open Lodge a claim	The message "You have no active policy" is shown and the Continue button is disabled.

CLM-102 – Enter the incident details

Scenario	Given	When	Then
1. Valid details	A policy is selected	The customer enters an incident date of today or earlier, the place, a claim type and a description	The details are accepted and the customer can go to the next step.
2. Validation failure (Negative)	The customer is on the incident details form	They enter an incident date in the future or leave the place empty	The field is highlighted with a message such as "Incident date cannot be in the future" and the Next button stays disabled.

CLM-103 – Show clear errors for missing or invalid fields

Scenario	Given	When	Then
1. Missing field (Negative)	The description field is empty	The customer clicks Submit	"Description is required" appears next to the field, the claim is not submitted and all other entries are kept.
2. Invalid date (Negative)	The incident date is set to tomorrow	The customer moves to the next field	The error "Incident date cannot be in the future" is shown next to the date field.
3. All fields valid	All required fields are filled with valid data	The customer reviews the form	No errors are shown and the Submit button is enabled.

CLM-104 – Submit the claim and receive a reference number

Scenario	Given	When	Then
1. Successful submission	A complete and valid claim form	The customer clicks Submit	The claim is saved with status Registered, a unique reference number (for example CLM-2026-000123) is shown and a confirmation is sent.
2. System unavailable (Negative)	A complete and valid form and the server is temporarily down	The customer clicks Submit	The message "We could not submit your claim. Your details are saved, please try again" is shown and no duplicate claim is created.

Task 5 – Planning Poker estimates and total backlog size

Reference story: CLM-301 Claims list with status = 3 points. It is well understood: one screen that lists the customer's claims from the database. Every other story is compared with it. Cards used: 1, 2, 3, 5, 8, 13.

ID	Story	Points	Reason (compared with CLM-301)
CLM-101	Select policy	3	A list of the customer's active policies plus a selection. Similar to 301.
CLM-102	Incident details form	3	One form with a few fields. Similar to 301.

ID	Story	Points	Reason (compared with CLM-301)
CLM-103	Field validation and errors	3	A few rules to check and error messages to show.
CLM-104	Submit and reference number	5	Saves the claim, generates a unique reference and sends a confirmation.
CLM-201	Upload a photo	5	File storage, type and size checks, and a document list.
CLM-204	Upload a PDF	2	Reuses the upload work of CLM-201, so only a new file type is added.
CLM-205	Block infected files	5	Uses a ready-made virus-scanning service. See the outlier discussion below.
CLM-202	Required documents checklist	3	A list of documents per claim type, with missing ones marked.
CLM-203	Officer verifies documents	5	A new officer screen with accept or reject, a reason and a saved result.
CLM-301	Claims list with status	3	**Reference story.**
CLM-302	Status history of a claim	3	A dated list of status changes for one claim.
CLM-303	Officer updates status	3	A status change with a saved history entry.
CLM-401	SMS and e-mail on decision	5	Two channels and two message templates, using an external provider.
CLM-402	Settlement amount and date	2	Displays two values that are already stored on the claim.
CLM-403	Approve or reject, record amount	5	An officer decision screen with rules and a saved amount.
CLM-404	Pay to bank and mark Settled	8	Bank integration, failure handling and a status change. See the outlier discussion below.
	Total backlog size	**63**	

Recorded discussion on the outliers

Story	Round 1 votes	Round 2 votes	Discussion
CLM-205 Block infected files	3, 5, 5, 8, 13	5, 5, 5, 5, 5	The voter with **13** thought the team would have to build a virus scanner. The voter with **3** had forgotten the rejection message for the customer. The team agreed to use a ready-made antivirus service and settled on **5**.
CLM-404 Pay to bank	3, 8, 8, 8, 13	8, 8, 8, 8, 8	The voter with **3** assumed the bank API is simple. The voter with **13** was worried about failed payments, retries and matching the payment to the claim. The bank offers a test API and a simple retry is enough, so the team settled on **8**.

The votes above are a sample of how the discussion is recorded.

Total backlog size

The total is **63 story points** across 16 stories. Per epic: E1 Claim Registration = 14, E2 Document Upload = 20, E3 Claim Tracking = 9, E4 Settlement Notifications = 20. If the team's velocity is about 20 points per sprint (an assumption, because no velocity was given), the backlog is about 3 to 4 sprints of work.

Estimated and ordered backlog

Rank	ID	Story	Epic	Points	Cumulative points
1	CLM-101	Select the policy	E1	3	3
2	CLM-102	Enter incident details	E1	3	6
3	CLM-103	Clear errors for invalid fields	E1	3	9
4	CLM-104	Submit claim, get reference	E1	5	14
5	CLM-201	Upload a photo	E2	5	19
6	CLM-204	Upload a PDF	E2	2	21
7	CLM-205	Block infected files	E2	5	26
8	CLM-202	Required documents checklist	E2	3	29
9	CLM-301	Claims list with status	E3	3	32
10	CLM-303	Officer updates status	E3	3	35
11	CLM-203	Officer verifies documents	E2	5	40
12	CLM-302	Status history of a claim	E3	3	43
13	CLM-403	Approve or reject, record amount	E4	5	48
14	CLM-402	Settlement amount and date	E4	2	50
15	CLM-404	Pay to bank, mark Settled	E4	8	58
16	CLM-401	SMS and e-mail on decision	E4	5	63
		Total		**63**	

Task 6 – JIRA project, board, epics, story states and DoD

How the project was set up

- A **Scrum** project named "ClaimEase" was created in Jira.
- The four epics were created: Claim Registration, Document Upload, Claim Tracking and Settlement Notifications.
- The 16 stories were created with their team ID in the title, linked to their epic, given story points and ordered in the Backlog by the rank in Task 5.
- The board has four story states: **To Do, In Progress, In Review and Done**.
- Sprint 1 holds the first five stories (19 points) with the Sprint Goal: "A policyholder can lodge a validated claim, receive a reference number and upload a first photo."
- The Definition of Done is created as a Jira task and attached as a file.

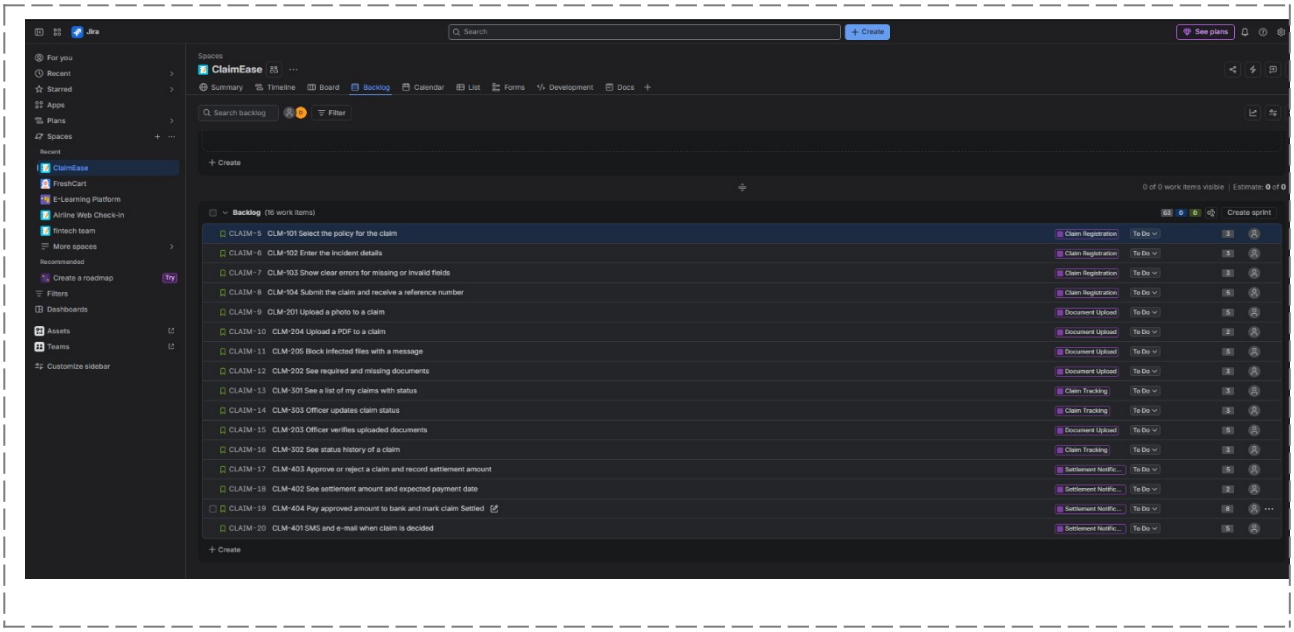


Figure 1: Jira Backlog view with epics, stories and story points

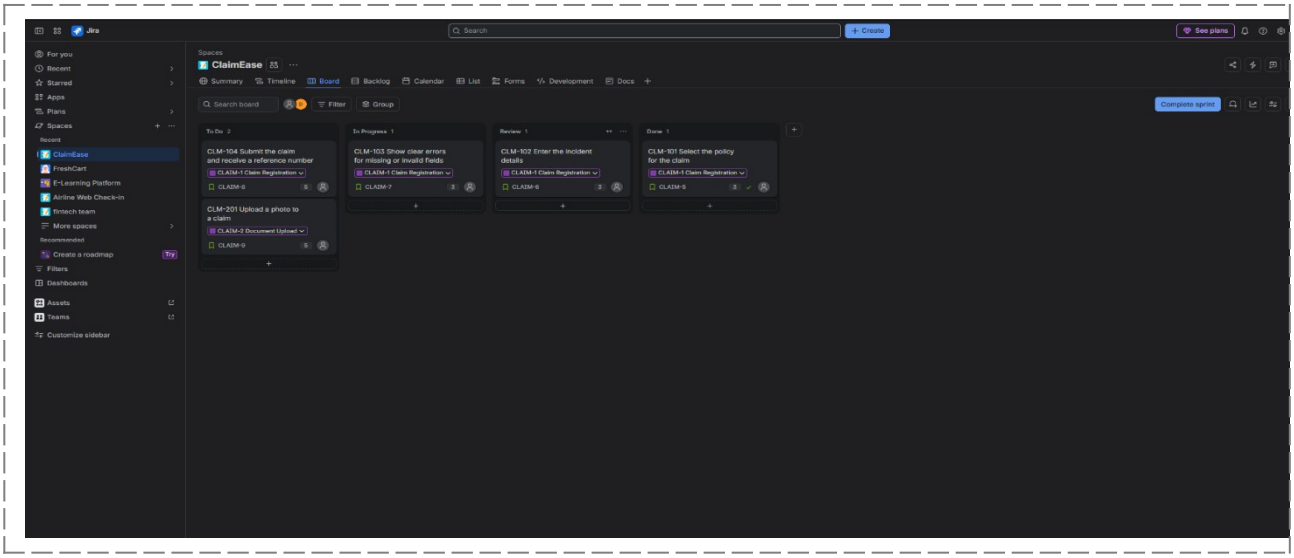


Figure 2.1: Jira board with the story states

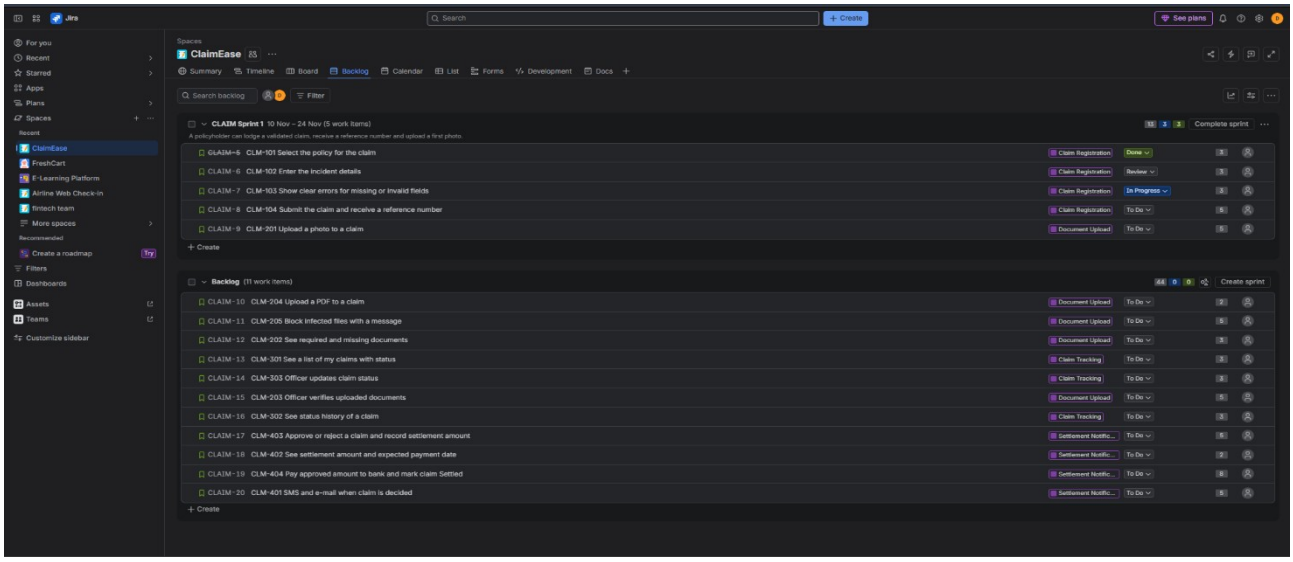


Figure 2.2: Jira board with the story states

Definition of Done (DoD)

A story is **done** only when all of these are true. The same list is attached to the Jira task "Definition of Done".

#	Criterion
1	The code is written, follows the coding standard and the pull request is reviewed and approved by a second developer.
2	Unit tests are written and pass, and new code has at least 80% test coverage.
3	Every acceptance criterion of the story is tested and passes, including the negative scenarios.
4	There are no open critical or major defects for the story.
5	The code is merged, built and deployed to the test environment, and the regression tests still pass.
6	Security and privacy are met: personal data and uploaded documents are stored securely and no sensitive data appears in logs.
7	Documentation is updated (API notes and on-screen help text).
8	The Product Owner has reviewed the story in a demo and accepted it. Only then does the card move to Done.

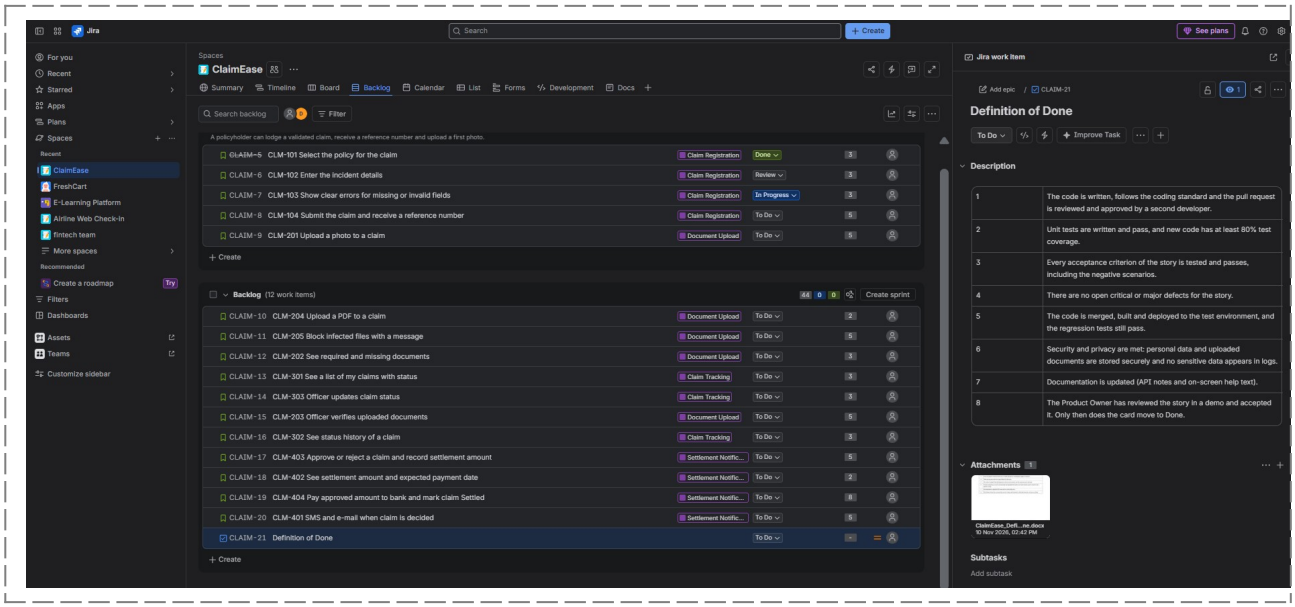


Figure 3: Definition of Done recorded and attached in Jira

Deliverable – One-page note on the roles, events and artifacts used

The ClaimEase programme uses Scrum with 2-week sprints. Jira is the tool that holds the backlog and the board.

Roles

Role	Who	What they do in this project
Product Owner	Head of Claims (assumed)	Owns the vision and the Product Backlog, orders the stories by value and accepts each story at the Sprint Review.
Scrum Master	One team member	Coaches the team in Scrum, runs the events, removes impediments such as a missing bank test account, and keeps the Jira board healthy.
Developers	Developers, testers and a UI designer	Estimate with Planning Poker, split stories vertically, and build and test each story to the Definition of Done.

Events

Event	Usual time-box	Purpose and use here
Sprint	2 weeks	The container for all other events. Sprint 1 aims to deliver claim registration and a first photo upload.
Sprint Planning	About 4 hours	The team chooses the top stories from the backlog and agrees the Sprint Goal.
Daily Scrum	15 minutes	Developers plan the next 24 hours and raise blockers, for example waiting for an API.
Sprint Review	About 2 hours	The team shows the working increment, and the Product Owner accepts stories and updates the backlog.
Sprint Retrospective	About 1.5 hours	The team improves its way of working, with one or two actions for the next sprint.

Backlog refinement (splitting and estimating stories) happens during the sprint. It is a regular activity and not a formal event.

Artifacts

Artifact	In this project	Commitment
Product Backlog	16 ordered stories under 4 epics, 63 story points, kept in Jira.	Product Goal (the product vision).
Sprint Backlog	The stories chosen for Sprint 1 and the plan to deliver them, shown on the board.	Sprint Goal.
Increment	The working, tested part of the portal that meets the Definition of Done.	Definition of Done.